

# Dhruv Roy Choudhary

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## SUMMARY

Electrical Engineering undergraduate at UC San Diego focused on machine learning, control, and autonomous systems. Experience in optimization, perception pipelines, and systems-level simulation across robotics and UAM routing applications.

## SKILLS & CERTIFICATIONS

**Certifications:** Machine Learning Specialization and Deep Learning Specialization (Stanford); Python for Data Science, AI, and Development (IBM)

**Skills:** Python, C++, MATLAB; **ML/Robotics:** PyTorch, OpenCV; **Systems:** Optimization, Embedded (Jetson, Raspberry Pi); **Tools:** Git, Cadence

## EDUCATION

- **University of California, San Diego** Sep 2023 – Mar 2027 (Expected)  
Bachelor of Sciences in Electrical and Electronics Engineering (Depth: Machine Learning and Controls) - 3.9 Major GPA San Diego, CA
  - Relevant Coursework: Probability & Statistics, Optimization, Circuits & Systems, Deep Learning, Pattern Recognition.

## EXPERIENCES

- **UC San Diego - RobotX Navigation Team (Advisor: Prof. Jack Silberman)** Apr 2026 - Present
  - Building UAV aerial perception and cross-domain communication pipeline for the Maritime RobotX 2026 Competition (Singapore, Nov 2026), targeting buoy detection and USV coordination
  - Developed hybrid HSV + YOLOv11n detection stack on Jetson Orin; trained on 110-image dataset achieving mAP50 0.967, precision 0.944, and F1 0.960 on held-out validation set
  - Engineering asynchronous ROS2 pub-sub architecture for UAV→USV GPS coordinate map transmission over RF/WiFi to enable cross-domain autonomous navigation
- **UC San Diego - Undergraduate Researcher (Advisor: Prof. Jorge Poveda)** Mar 2026 - Present
  - Developing multi-agent game-theoretic simulation frameworks for model-free Nash equilibrium seeking under adversarial perturbations and strategic behavior
- **NeuralSeek** Jan 2026 - Mar 2026  
Agentic AI Intern Miami, FL
  - Created [FaultWise](#), an agentic circuit/PCB validation tool combining rule-based electrical reasoning with AI-driven constraint checks to detect unsafe configurations and recommend fixes before physical prototyping.
- **FastFlyrr LLC ([Non-Confidential Repository](#))** Oct 2025 - Mar 2026  
AI Navigators Team Intern Tampa, FL
  - Designed and implemented day-ahead UAM route scheduler using time-expanded conflict graphs and greedy Maximum Weighted Independent Set (MWIS) to generate conflict-free flight plans under weather, delay, and vertiport constraints.
  - Built scheduler-validator integration supporting multi-refuel routing and release-time awareness, eliminating infeasible plans through global consistency checks
- **Qualcomm Institute (Advisor: Dr. Prasad Gudem)** Sep 2024 - Sep 2025  
Research Apprentice San Diego, CA
  - Built a block-level 5G transceiver prototype for graduate-level instruction and demonstration at G.E.A.R 2025 Symposium.
- **TritonBot FC ([Team Description Paper](#))** Sep 2024 - June 2025  
UC San Diego RoboCup SSL Mechatronics Team San Diego, CA
  - Engineered LT3751-based flyback transformer systems for solenoid actuation for autonomous robots.

## PROJECTS

- **Robust Semantic Segmentation under Distribution Shift ([Report](#)) ([Repository](#))** Jan 2026 - Mar 2026
  - Built DeepLabV3-ResNet50 pipeline on Cityscapes with synthetic corruptions (blur, noise, brightness, JPEG) to evaluate robustness under distribution shift
  - Designed confidence-guided selective repair pipeline (thresholding + region fill + diffusion) improving corrupted mIoU by  $\sim +0.04$  and reducing pixel error by  $\sim 12.7\%$  on edited regions
- **CARLA-Based Multi-Object Tracking Stability Analysis ([Repository](#))** Dec 2025 - Jan 2026
  - Built an end-to-end CARLA perception pipeline (RGB  $\rightarrow$  YOLOv8  $\rightarrow$  ByteTrack / BoT-SORT), exporting MoT-style labels over  $\sim 4.5k$  frames to analyze detection-tracking coupling.
  - Quantified the precision-continuity trade-off in online MOT: increasing detector confidence (0.40 $\rightarrow$ 0.65) reduced active tracks/frame by  $\sim 60\%$  while increasing  $<1s$  track churn by  $\sim 38\%$ , showing ID stability is dominated by detection sparsity.
  - Generated trajectory dataset ( $\sim 2k$  sequences) and benchmarked CV, Kalman, and GRU models.

## LEADERSHIP AND AWARDS

- **IEEE UCSD: Social Chair; 2024 Quarterly Projects Winner** (Physical RGB Clock with circadian rhythm based lighting)
- **Pre-university:** Founded **Engineering Society** and **Project Eureka!** (non-profit) to expand STEM access in Kuala Lumpur.